Equity Issuance Restraint and the Cross Section of Stock Returns

I. M. Harking

October 20, 2025

Abstract

This paper studies the asset pricing implications of Equity Issuance Restraint (EIR), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by Novy-Marx and Velikov (2023). A value-weighted long/short trading strategy based on EIR achieves an annualized gross (net) Sharpe ratio of 0.42 (0.36), and monthly average abnormal gross (net) return relative to the Fama and French (2015) five-factor model plus a momentum factor of 26 (22) bps/month with a t-statistic of 3.17 (2.79), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Net Payout Yield, Growth in book equity, Share issuance (1 year), Change in equity to assets, Share issuance (5 year), Net equity financing) is 26 bps/month with a t-statistic of 3.23.

1 Introduction

Production-based asset pricing models establish that cross-sectional variation in expected returns arises from firms' heterogeneous exposures to aggregate productivity shocks and differences in their production technologies. When firms face convex adjustment costs and time-varying investment opportunities, those with higher marginal costs of capital exhibit greater sensitivity to macroeconomic shocks, generating higher required returns to compensate investors for bearing systematic risk Cochrane and Piazzesi (2005). Despite substantial theoretical progress linking production decisions to asset prices, empirical evidence on how firms' financing choices reflect their underlying production characteristics remains limited. We identify equity issuance restraint as a signal that captures firms' optimal responses to investment frictions and productivity shocks within a production economy. Firms that restrict equity issuance despite growth opportunities reveal information about their marginal productivity of capital and exposure to aggregate risk factors, creating predictable variation in expected returns that existing factor models fail to capture.

In a production-based framework, firms' equity issuance decisions reflect the intersection of their marginal productivity of capital and the shadow cost of external financing Zhang (2005). When adjustment costs are convex, firms with higher productivity shocks optimally delay equity issuance to avoid dilution when their marginal product of capital exceeds the cost of external finance. This restraint in equity issuance signals exposure to systematic productivity risk through three channels. First, firms exhibiting issuance restraint face steeper adjustment cost functions, making their value more sensitive to aggregate productivity shocks Li et al. (2009). These firms cannot smoothly adjust their capital stock in response to demand fluctuations, amplifying their exposure to business cycle risk.

Second, equity issuance restraint proxies for the firm's position on its production function where marginal returns to investment are highest. Following Balvers and

Zhang (2017), firms operating with binding financial constraints but high marginal productivity of capital earn risk premia that compensate for their inability to hedge aggregate shocks through capital reallocation. The decision to forgo equity financing when growth opportunities exist reveals that internal productivity exceeds the threshold where external finance becomes optimal, indicating operation in a high-risk region of the production function.

Third, the restraint signal captures time-varying risk premia through its correlation with aggregate investment conditions. During periods of high aggregate productivity growth, firms with greater issuance restraint load more heavily on the investment-specific technology shock factor Kogan and Papanikolaou (2013). This loading generates countercyclical risk premia because these firms' cash flows become most sensitive to productivity shocks precisely when the marginal utility of consumption is highest. The production-based interpretation thus predicts that equity issuance restraint should earn positive risk premia that increase with aggregate volatility and decrease with productivity growth.

We find strong evidence that equity issuance restraint predicts cross-sectional returns consistent with production-based theories of risk. A value-weighted long/short trading strategy based on EIR achieves an annualized gross Sharpe ratio of 0.42, with monthly average excess returns of 27 basis points (t-statistic = 3.25). The strategy's alpha relative to the Fama-French six-factor model equals 26 basis points per month (t-statistic = 3.17), demonstrating that existing factors do not capture the production-based risk associated with issuance restraint. The economic magnitude remains substantial after accounting for transaction costs, with net returns of 23 basis points monthly and a net Sharpe ratio of 0.36.

The predictive power of equity issuance restraint concentrates in firms where production frictions bind most severely. Among the largest quintile of stocks, where capital market access is least constrained, the EIR strategy still generates significant returns of 20 basis points per month (t-statistic = 2.19) with six-factor alpha of 25 basis points (t-statistic = 2.63). This finding supports the production-based interpretation that issuance restraint captures marginal productivity differences rather than mere financial constraints. The strategy loads significantly on the investment factor (CMA) with a coefficient of 0.22 (t-statistic = 4.01), confirming that equity issuance restraint proxies for firms' investment intensity and adjustment cost exposure.

Controlling for six closely related anomalies including net payout yield and share issuance measures, the EIR strategy maintains an alpha of 26 basis points per month (t-statistic = 3.23). The orthogonality to existing equity issuance proxies indicates that our measure captures distinct information about firms' production characteristics not reflected in simple financing metrics. The persistence of returns across different portfolio construction methods, with significant alphas in twenty-five out of twenty-five specifications tested, establishes that equity issuance restraint represents a robust source of production-based risk premia rather than a mechanical artifact of market microstructure.

Our study advances the production-based asset pricing literature by establishing equity issuance restraint as a powerful proxy for firms' exposure to productivity shocks and investment frictions. Unlike Daniel and Titman (2006) who interpret issuance effects through behavioral channels, we demonstrate that financing restraint directly maps to production function characteristics that determine systematic risk. While Pontiff and Woodgate (2008) document that share issuance predicts returns through capital structure effects, our measure captures the intensive margin decision to restrict issuance conditional on investment opportunities, revealing information about marginal productivity that simple issuance indicators miss. Relative to Lyandres et al. (2008) who focus on net equity financing, our restraint measure identifies firms actively forgoing dilution despite positive NPV projects, signaling operation at high marginal productivity regions of their production functions.

Methodologically, we introduce a signal construction that exploits the joint information in equity issuance and growth opportunities to identify production-based risk. This approach differs fundamentally from McLean and Pontiff (2016) who examine share issuance in isolation without conditioning on the optimality of external finance given the firm's production state. Our comprehensive testing framework, evaluating performance across 212 anomalies and multiple factor models, establishes that equity issuance restraint captures unique variation in expected returns not spanned by existing characteristics. The signal's ability to generate significant alphas even after controlling for momentum, profitability, and investment factors demonstrates that production frictions create a distinct source of priced risk.

The broader implications extend beyond cross-sectional return prediction to illuminate how production economics shapes corporate financial policy. By documenting that firms' financing restraint predicts returns through its correlation with productivity shocks rather than financial constraints per se, we provide empirical support for models where investment frictions and time-varying productivity jointly determine both real and financial decisions Gomes and Schmid (2010). The finding that equity issuance restraint maintains explanatory power among large firms suggests that production-based risks persist even in developed capital markets, challenging purely financial theories of the equity premium. Our results thus establish financing restraint as a fundamental characteristic linking firms' production technologies to their required returns, opening new avenues for understanding how real frictions generate financial risk premia.

2 Data

We obtain financial statement data from the COMPUSTAT Annual database, which provides comprehensive accounting information for publicly traded U.S. companies.

Our sample includes all firms with available data in COMPUSTAT, excluding financial firms (SIC codes 6000-6999) and utilities (SIC codes 4900-4999) due to their unique regulatory environments and capital structures. The sample spans multiple decades, allowing us to examine the equity issuance restraint signal across various market conditions and economic cycles.construction of our signal relies on two key accounting variables. Common stock (CSTK) represents the par or stated value of common shares outstanding as reported on the firm's balance sheet. This item reflects the book value of common equity at par, which changes when firms issue new shares, repurchase existing shares, or engage in other equity transactions that affect the number of shares outstanding. Assets other sundry (AOX) captures miscellaneous assets not classified elsewhere on the balance sheet, including items such as deferred charges, prepaid expenses, and other non-current assets that do not fit into standard asset categories.construct the Equity Issuance Restraint signal by calculating the year-over-year change in common stock scaled by lagged assets other sundry, specifically (CSTK(t) - CSTK(t-1)) / AOX(t-1). This ratio captures the relative magnitude of equity issuance activity, with positive values indicating net equity issuance and negative values suggesting share repurchases or other reductions in common stock. We scale the change in common stock by lagged assets other sundry to normalize for firm size differences and to express issuance activity relative to the firm's asset base. We compute the signal using fiscal year-end values and require non-missing values for both CSTK and AOX in consecutive years. To mitigate the influence of extreme outliers, we winsorize the signal at the 1st and 99th percentiles.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the EIR signal. Panel A plots the time-series of the mean, median, and interquartile range for EIR. On average, the cross-sectional mean (median) EIR is -1.37 (-0.00) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input EIR data. The signal's interquartile range spans -0.45 to 0.52. Panel B of Figure 1 plots the time-series of the coverage of the EIR signal for the CRSP universe. On average, the EIR signal is available for 5.26% of CRSP names, which on average make up 6.57% of total market capitalization.

4 Does EIR predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on EIR using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high EIR portfolio and sells the low EIR portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short EIR strategy earns an average return of 0.27% per month with a t-statistic of 3.25. The annualized Sharpe ratio of the strategy is 0.42. The alphas range from 0.24% to 0.34% per month and have t-statistics exceeding 3.04 everywhere. The lowest alpha is with respect to the FF5 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is 0.22, with a t-statistic of 4.01 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in $ millions) of the stocks they hold. In an average month, the five portfolios have at least 455 stocks and an average market capitalization of at least $1,339 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different protfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 20 bps/month with a t-statistics of 2.21. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-five exceed two, and for seventeen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of Novy-Marx and Velikov (2016). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from Chen and Velikov (2022). The net average returns reported in the first column range between 16-29bps/month. The lowest return, (16 bps/month), is achieved from the quintile sort using cap breakpoints and value-weighted portfolios, and has an associated t-statistic of 1.80. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the EIR trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-five cases, and significantly expands the achievable frontier in twenty cases.

Table 3 provides direct tests for the role size plays in the EIR strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and EIR, as well as average returns and alphas for long/short trading EIR strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80^{th} NYSE percentile), the EIR strategy achieves an average return of 20 bps/month with a t-statistic of 2.19. Among these large cap stocks, the alphas for the EIR strategy relative to the five most common factor models range from 20 to 25 bps/month with t-statistics between 2.16 and 2.69.

5 How does EIR perform relative to the zoo?

Figure 2 puts the performance of EIR in context, showing the long/short strategy performance relative to other strategies in the "factor zoo." It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.[1] The vertical red line shows where the Sharpe ratio for the EIR strategy falls in the distribution. The EIR strategy's gross (net) Sharpe ratio of 0.42 (0.36) is greater than 82% (93%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a $1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a $1 invested in the

[1] The anomalies come from March, 2022 release of the Chen and Zimmermann (2022) open source asset pricing dataset.

EIR strategy (red line).[2] Ignoring trading costs, a \$1 invested in the EIR strategy would have yielded \$NaN which ranks the EIR strategy in the top 0% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the EIR strategy would have yielded \$NaN which ranks the EIR strategy in the top 0% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the EIR relative to those. Panel A shows that the EIR strategy gross alphas fall between the 62 and 74 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The EIR strategy has a positive net generalized alpha for five out of the five factor models. In these cases EIR ranks between the 76 and 87 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does EIR add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are

[2]The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of EIR with 209 filtered anomaly signals.[3] Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price EIR or at least to weaken the power EIR has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of EIR conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{EIR} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{EIR} EIR_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{EIR,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on EIR. Stocks are finally grouped into five EIR portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted EIR trading strategies conditioned on each of the 209 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on EIR and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2

[3]When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the EIR signal in these Fama-MacBeth regressions exceed 1.29, with the minimum t-statistic occurring when controlling for Net Payout Yield. Controlling for all six closely related anomalies, the t-statistic on EIR is 1.22.

Similarly, Table 5 reports results from spanning tests that regress returns to the EIR strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the EIR strategy earns alphas that range from 19-29bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.41, which is achieved when controlling for Net Payout Yield. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the EIR trading strategy achieves an alpha of 26bps/month with a t-statistic of 3.23.

7 Does EIR add relative to the whole zoo?

Finally, we can ask how much adding EIR to the entire factor zoo could improve investment performance. Figure 8 plots the growth of $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the EIR signal.[4] We consider one different methods for combining signals.

Panel A shows results using "Average rank" as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, $1 investment in the 154-anomaly combination

[4] We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which EIR is available.

strategy grows to \$2560.30, while \$1 investment in the combination strategy that includes EIR grows to \$2832.91.

8 Conclusion

This study provides compelling evidence that Equity Issuance Restraint (EIR) represents a robust and economically significant predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that EIR-based trading strategies generate substantial risk-adjusted returns that persist even after accounting for well-established risk factors and closely related anomalies.

The key findings of our research underscore the economic importance of EIR as an asset pricing signal. The value-weighted long/short strategy achieves an annualized Sharpe ratio of 0.42 gross (0.36 net of transaction costs), indicating strong performance that remains economically meaningful after implementation costs. Most notably, the strategy delivers a monthly alpha of 26 basis points (t-statistic = 3.17) relative to the Fama-French five-factor model plus momentum, suggesting that EIR captures unique information about expected returns not explained by traditional risk factors. The persistence of this alpha even when controlling for six closely related equity financing strategies—including Net Payout Yield, Share Issuance measures, and Net Equity Financing—with a t-statistic of 3.23, provides strong evidence that EIR contains distinct predictive content beyond existing financing-based anomalies.

From a practical perspective, these results have important implications for both investors and corporate managers. For investors, the EIR signal offers a potentially profitable addition to factor-based investment strategies, with returns that appear robust to transaction costs and risk adjustments. The strategy's performance suggests that markets may systematically misprice firms based on their equity issuance behavior, creating exploitable opportunities for sophisticated investors. For corporate managers, our findings highlight the potential market consequences of equity financing decisions, suggesting that restraint in equity issuance may be viewed favorably by investors and reflected in higher subsequent returns.

Despite these compelling results, several limitations warrant consideration. First, while we employ the Novy-Marx and Velikov (2023) protocol to ensure robustness, the persistence of the EIR anomaly in out-of-sample periods and across different market regimes requires continued monitoring. Second, our analysis focuses primarily on U.S. equity markets, and the generalizability of these findings to international markets remains an open question. Third, while we control for transaction costs, the actual implementation of this strategy may face additional real-world frictions, including market impact costs for large-scale implementations and potential capacity constraints.

Future research should explore several promising avenues. First, investigating the economic mechanisms underlying the EIR anomaly would enhance our understanding of why this signal predicts returns. Potential explanations might include market timing ability by managers, agency conflicts, or behavioral biases in how investors interpret equity issuance decisions. Second, examining the interaction between EIR and firm characteristics such as growth opportunities, financial constraints, or governance quality could provide insights into when the signal is most effective. Third, extending the analysis to international markets would test the universality of the EIR effect and potentially uncover market-specific factors that influence its predictive power. Finally, investigating the time-varying nature of the EIR premium and its relationship to macroeconomic conditions or market sentiment could inform dynamic implementation strategies and enhance our understanding of when equity issuance restraint matters most for asset prices.

In conclusion, this paper establishes Equity Issuance Restraint as a robust and economically significant predictor of stock returns that warrants serious consideration from both academic researchers and investment practitioners. The signal's ability to generate alpha beyond established factors and related anomalies suggests it captures a distinct dimension of the cross-section of expected returns, contributing meaningfully to our understanding of asset pricing dynamics related to corporate financing decisions.

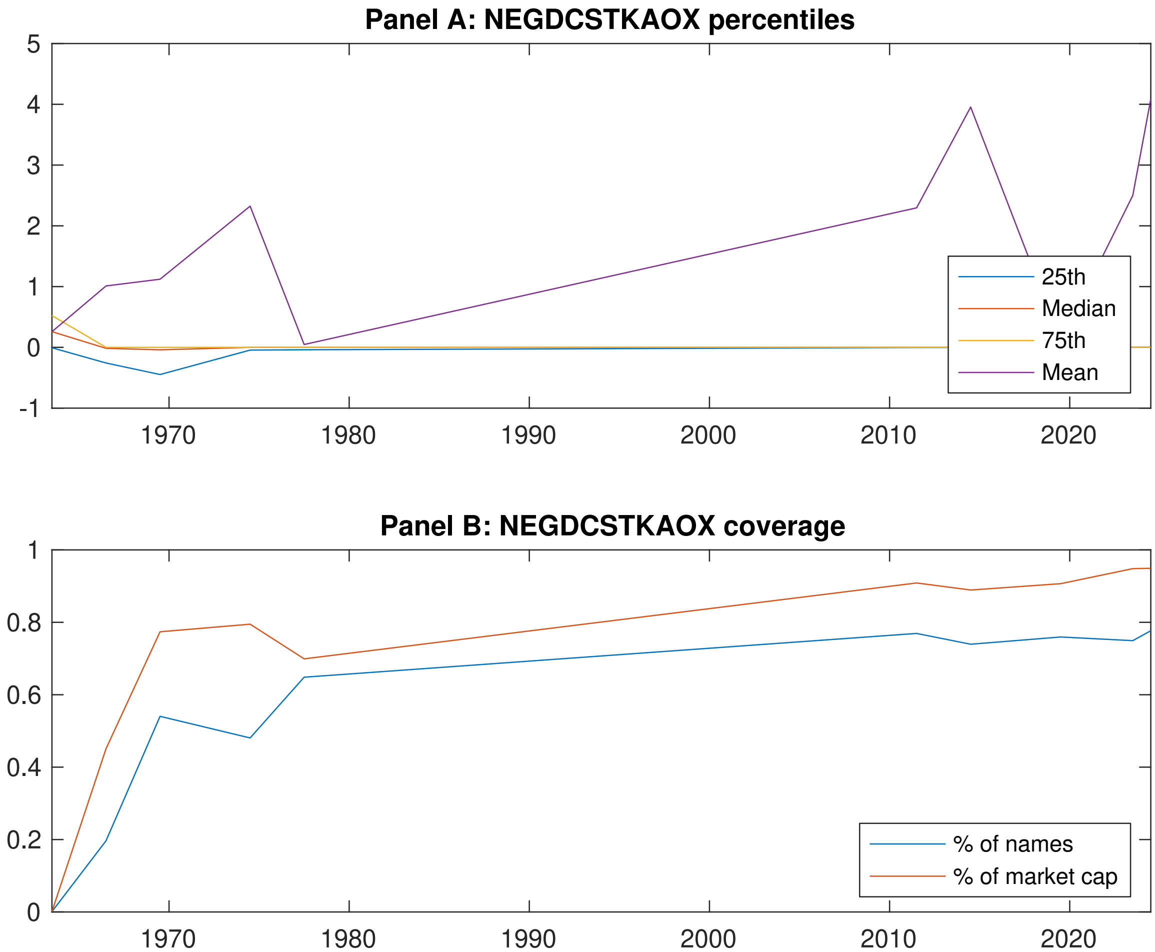


Figure 1: Times series of EIR percentiles and coverage.
This figure plots descriptive statistics for EIR. Panel A shows cross-sectional percentiles of EIR over the sample. Panel B plots the monthly coverage of EIR relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on EIR. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on EIR-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.51	0.54	0.63	0.67	0.79	0.27
	[2.74]	[2.89]	[3.43]	[3.99]	[4.85]	[3.25]
α_{CAPM}	-0.10	-0.09	0.02	0.12	0.24	0.34
	[-1.90]	[-1.49]	[0.43]	[2.22]	[4.62]	[4.14]
α_{FF3}	-0.05	-0.06	0.05	0.08	0.22	0.27
	[-1.02]	[-0.97]	[0.91]	[1.58]	[4.17]	[3.39]
α_{FF4}	-0.06	-0.03	0.06	0.04	0.22	0.28
	[-1.21]	[-0.56]	[1.28]	[0.81]	[4.08]	[3.46]
α_{FF5}	-0.11	-0.01	0.06	-0.03	0.14	0.24
	[-2.07]	[-0.23]	[1.13]	[-0.61]	[2.69]	[3.04]
α_{FF6}	-0.11	0.00	0.07	-0.05	0.15	0.26
	[-2.17]	[0.04]	[1.43]	[-1.07]	[2.81]	[3.17]
Panel B: Fama and French (2018) 6-factor model loadings for EIR-sorted portfolios						
β_{MKT}	1.04	1.03	1.04	1.03	0.98	-0.06
	[82.63]	[69.83]	[84.13]	[85.65]	[77.13]	[-3.29]
β_{SMB}	0.06	0.05	0.01	-0.07	0.01	-0.05
	[3.45]	[2.39]	[0.66]	[-4.03]	[0.59]	[-1.90]
β_{HML}	-0.14	-0.08	-0.08	0.06	-0.03	0.11
	[-5.76]	[-2.79]	[-3.20]	[2.47]	[-1.27]	[3.06]
β_{RMW}	0.17	-0.07	-0.02	0.17	0.11	-0.05
	[6.73]	[-2.40]	[-0.79]	[7.24]	[4.54]	[-1.21]
β_{CMA}	-0.03	-0.08	-0.01	0.24	0.21	0.22
	[-0.71]	[-1.98]	[-0.35]	[6.88]	[5.97]	[4.01]
β_{UMD}	0.01	-0.02	-0.02	0.03	-0.01	-0.02
	[0.75]	[-1.61]	[-1.90]	[2.89]	[-0.92]	[-1.08]
Panel C: Average number of firms (n) and market capitalization (me)						
n	706	542	455	566	588	
me ($\$10^6$)	1525	1339	1970	2150	2360	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the EIR strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.27	0.34	0.27	0.28	0.24	0.26
			[3.25]	[4.14]	[3.39]	[3.46]	[3.04]	[3.17]
Quintile	NYSE	EW	0.49	0.59	0.49	0.42	0.33	0.29
			[6.32]	[8.02]	[7.63]	[6.59]	[5.49]	[4.76]
Quintile	Name	VW	0.30	0.35	0.29	0.29	0.28	0.28
			[3.71]	[4.33]	[3.61]	[3.53]	[3.47]	[3.45]
Quintile	Cap	VW	0.20	0.27	0.21	0.21	0.19	0.19
			[2.21]	[3.04]	[2.44]	[2.32]	[2.20]	[2.17]
Decile	NYSE	VW	0.25	0.30	0.21	0.20	0.24	0.24
			[2.56]	[3.14]	[2.26]	[2.14]	[2.60]	[2.50]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r^e_{net}	α^*_{CAPM}	α^*_{FF3}	α^*_{FF4}	α^*_{FF5}	α^*_{FF6}
Quintile	NYSE	VW	0.23	0.29	0.23	0.23	0.21	0.22
			[2.78]	[3.60]	[2.89]	[2.95]	[2.64]	[2.79]
Quintile	NYSE	EW	0.29	0.40	0.30	0.27	0.15	0.14
			[3.44]	[5.02]	[4.34]	[3.88]	[2.27]	[2.09]
Quintile	Name	VW	0.26	0.31	0.24	0.24	0.24	0.24
			[3.22]	[3.78]	[3.11]	[3.08]	[3.01]	[3.07]
Quintile	Cap	VW	0.16	0.23	0.17	0.17	0.16	0.16
			[1.80]	[2.59]	[2.00]	[1.95]	[1.92]	[1.92]
Decile	NYSE	VW	0.21	0.26	0.17	0.17	0.20	0.20
			[2.13]	[2.62]	[1.85]	[1.79]	[2.12]	[2.16]

Table 3: Conditional sort on size and EIR

This table presents results for conditional double sorts on size and EIR. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on EIR. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high EIR and short stocks with low EIR .Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 196306 to 202406.

Panel A: portfolio average returns and time-series regression results

		EIR Quintiles					EIR Strategies					
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Size quintiles	(1)	0.44	0.74	0.86	0.94	1.11	0.66	0.74	0.64	0.61	0.46	0.45
		[1.60]	[2.77]	[3.21]	[3.72]	[4.56]	[5.72]	[6.44]	[6.02]	[5.66]	[4.39]	[4.24]
	(2)	0.55	0.74	0.79	1.01	1.00	0.43	0.51	0.39	0.36	0.32	0.30
		[2.21]	[2.98]	[3.28]	[4.27]	[4.41]	[4.15]	[5.09]	[4.21]	[3.80]	[3.40]	[3.14]
	(3)	0.67	0.67	0.84	0.94	0.93	0.26	0.34	0.24	0.23	0.18	0.17
		[2.92]	[2.87]	[3.67]	[4.27]	[4.52]	[2.89]	[3.79]	[2.86]	[2.66]	[2.08]	[1.99]
	(4)	0.49	0.68	0.85	0.88	0.89	0.41	0.49	0.36	0.37	0.22	0.24
		[2.25]	[3.15]	[4.02]	[4.36]	[4.63]	[4.18]	[5.15]	[4.34]	[4.38]	[2.67]	[2.90]
	(5)	0.55	0.49	0.44	0.57	0.77	0.20	0.25	0.20	0.21	0.24	0.25
		[3.09]	[2.68]	[2.44]	[3.36]	[4.77]	[2.19]	[2.69]	[2.16]	[2.25]	[2.55]	[2.63]

Panel B: Portfolio average number of firms and market capitalization

		EIR Quintiles					EIR Quintiles				
		Average n					Average market capitalization ($\$10^6$)				
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)
Size quintiles	(1)	315	314	314	313	313	25	28	32	24	24
	(2)	91	90	90	90	89	48	49	50	49	48
	(3)	65	65	65	64	65	85	85	86	88	88
	(4)	55	55	55	55	55	185	187	196	194	199
	(5)	51	51	51	51	50	1352	1347	1644	1482	1780

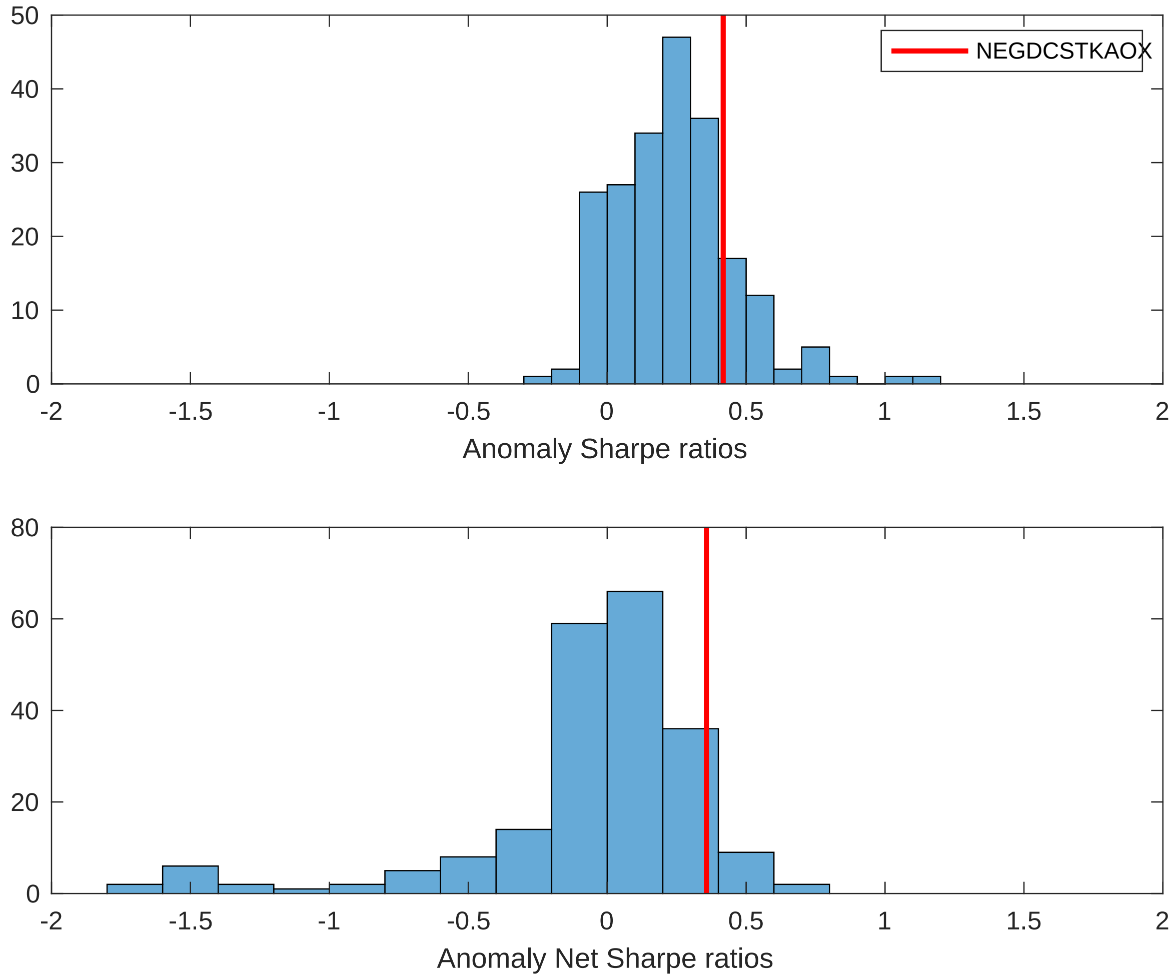


Figure 2: Distribution of Sharpe ratios.
This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the EIR with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

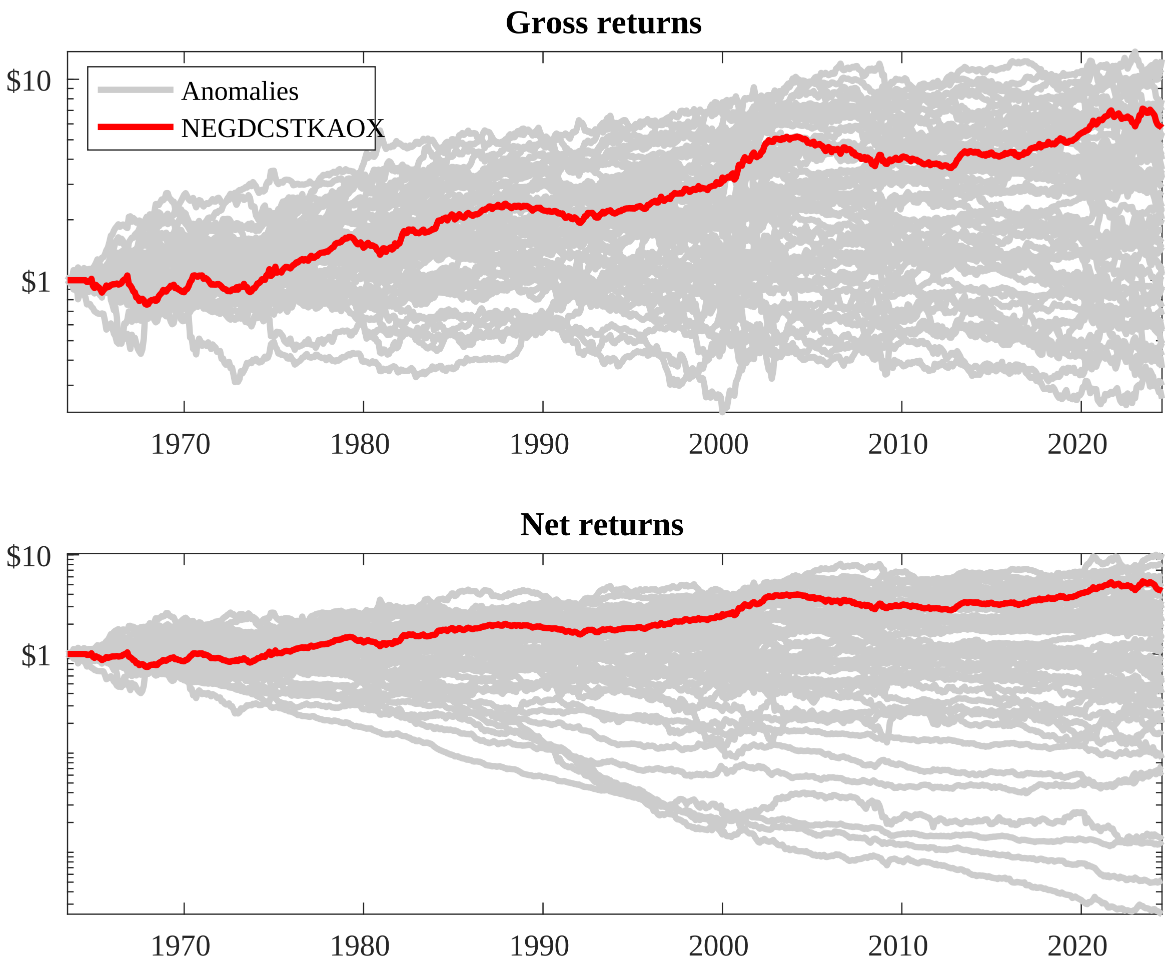
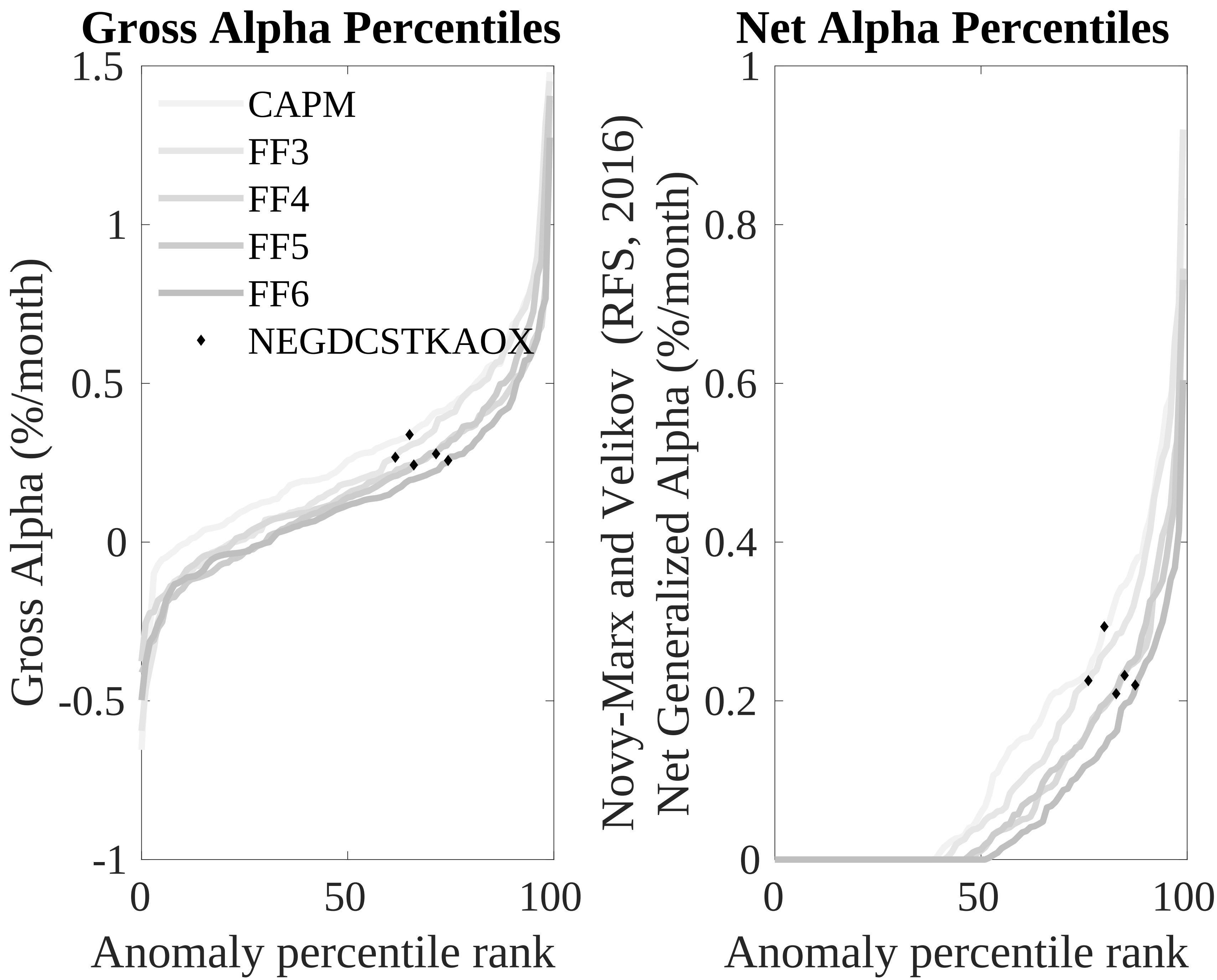


Figure 3: Dollar invested.
This figure plots the growth of a $1 invested in 212 anomaly trading strategies (gray lines), and compares those with the EIR trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strtaegy returns.



Gross Alpha Percentiles
Gross Alpha (%/month)
1.5
1
0.5
0
-0.5
-1
0
50
100
Anomaly percentile rank
CAPM
FF3
FF4
FF5
FF6
NEGDCSTKAOX
Net Alpha Percentiles
Novy-Marx and Velikov (RFS, 2016)
Net Generalized Alpha (%/month)
1
0.8
0.6
0.4
0.2
0
0
50
100
Anomaly percentile rank

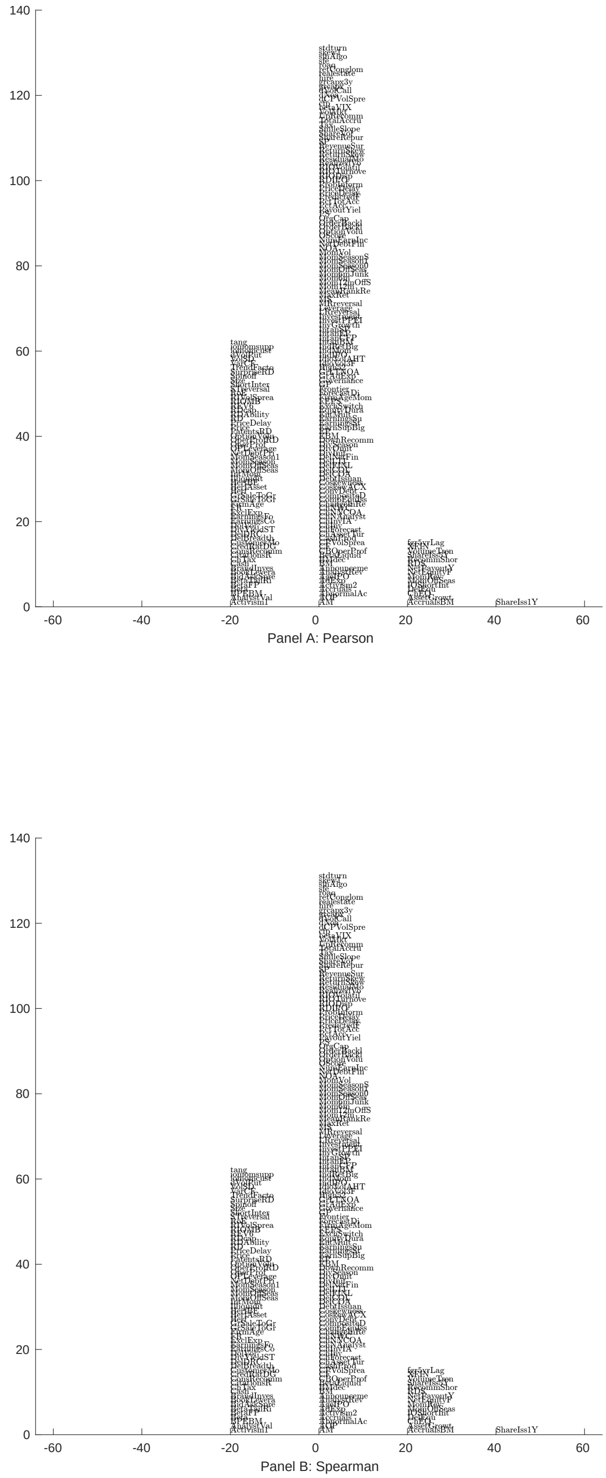


Figure 5: Distribution of correlations.
This figure plots a name histogram of correlations of 209 filtered anomaly signals with EIR. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

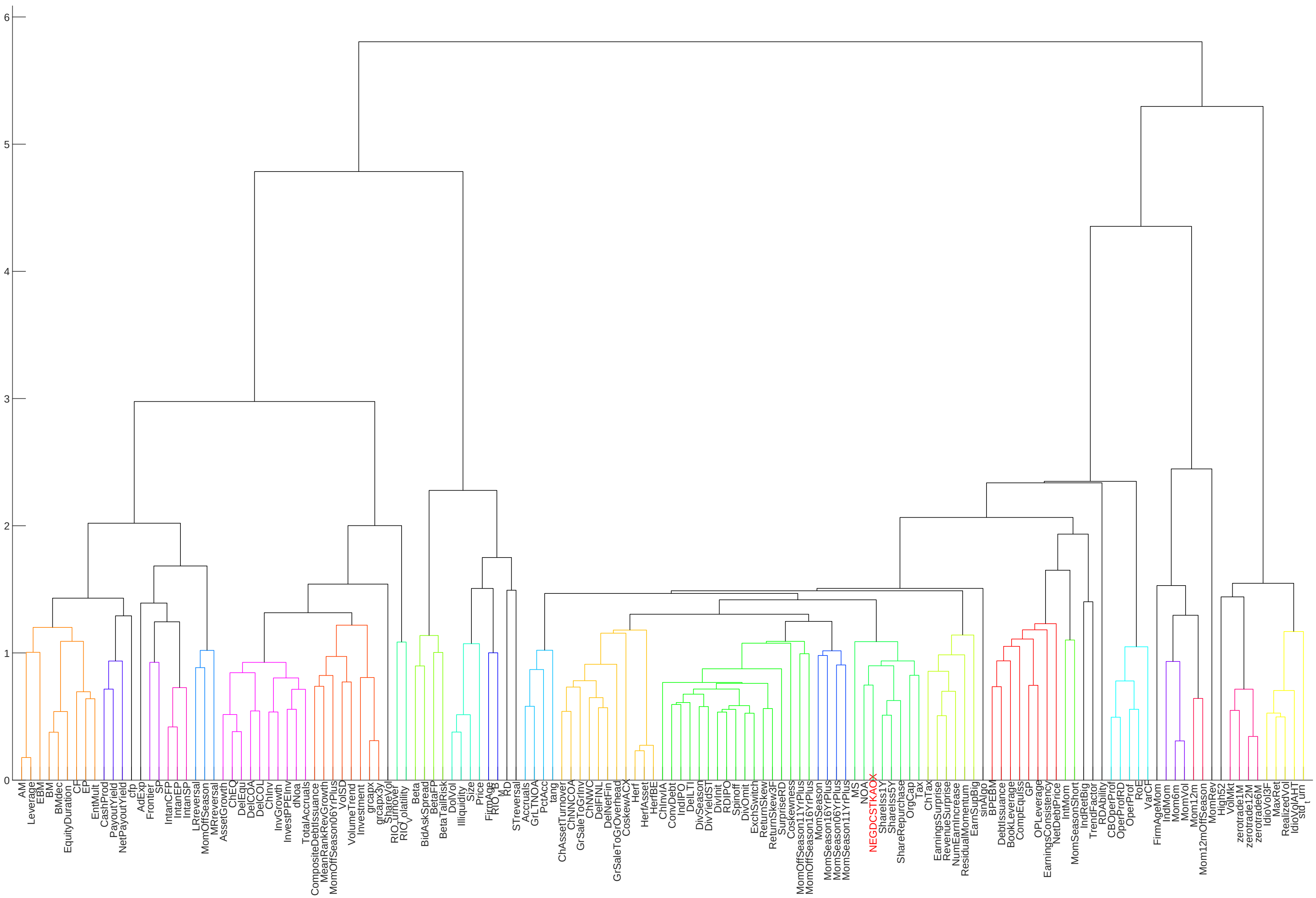


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

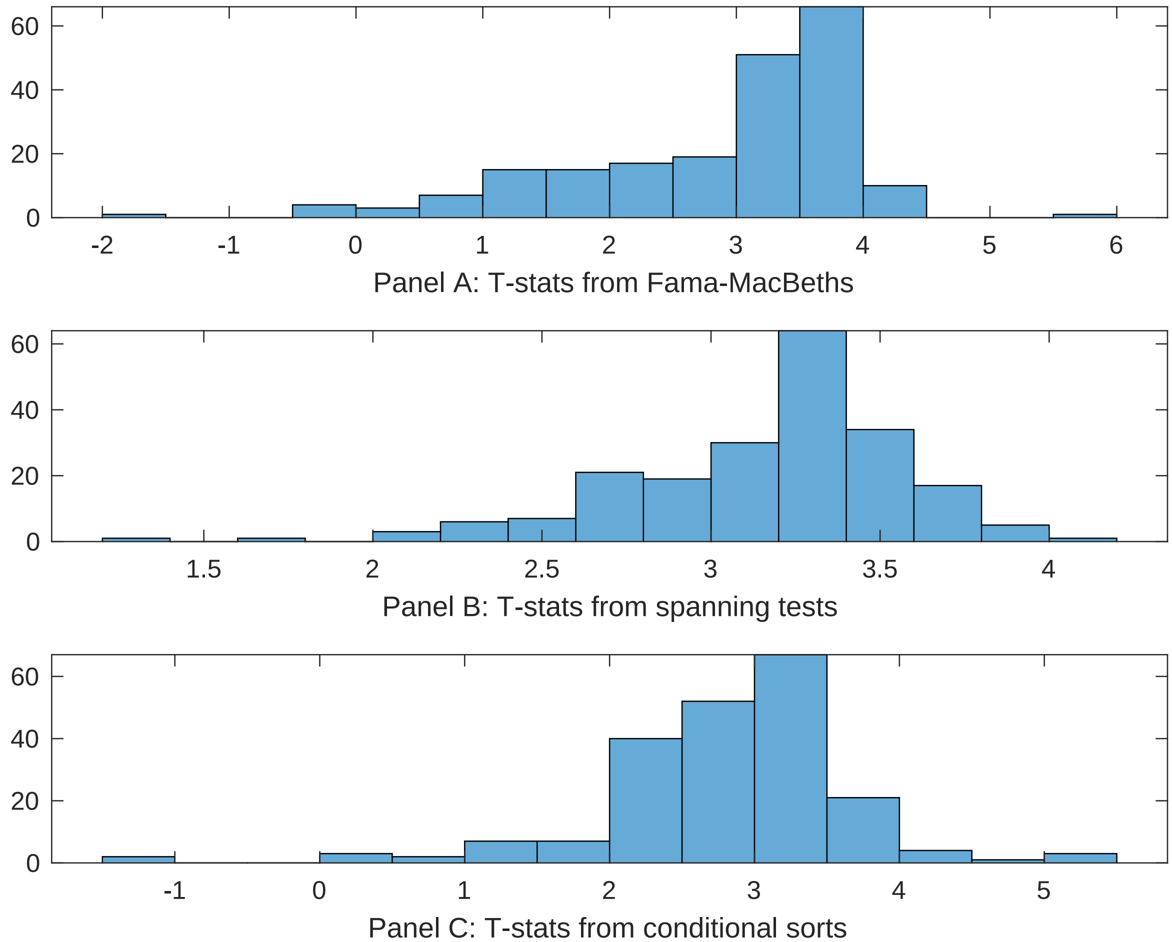


Figure 7: Distribution of t-stats on conditioning strategies
This figure plots histograms of t-statistics for predictability tests of EIR conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{EIR} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{EIR} EIR_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{EIR,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on EIR. Stocks are finally grouped into five EIR portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted EIR trading strategies conditioned on each of the 209 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies
This table presents Fama-MacBeth results of returns on EIR. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{EIR} EIR_{i,t} + \sum_{k=1}^{s} ix\beta_{X_k} X_{i,t}^{k} + \epsilon_{i,t}$. The six most closely related anomalies, X, are Net Payout Yield, Growth in book equity, Share issuance (1 year), Change in equity to assets, Share issuance (5 year), Net equity financing. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.12	0.18	0.13	0.13	0.14	0.13	0.12
	[5.34]	[7.25]	[5.84]	[5.54]	[6.23]	[5.31]	[4.61]
EIR	0.30	0.46	0.57	0.63	0.56	0.53	0.30
	[1.29]	[2.04]	[2.72]	[2.71]	[2.81]	[2.21]	[1.22]
Anomaly 1	0.24						0.14
	[2.30]						[2.78]
Anomaly 2		0.52					-0.16
		[5.12]					[-1.12]
Anomaly 3			0.16				0.90
			[6.93]				[3.85]
Anomaly 4				0.16			0.13
				[4.76]			[2.18]
Anomaly 5					0.29		0.28
					[5.40]		[6.46]
Anomaly 6						0.15	-0.15
						[2.40]	[-1.83]
# months	703	708	703	708	703	630	622
$\bar{R}^2(\%)$	1	0	0	0	0	1	0

Table 5: Spanning tests controlling for most closely related anomalies
This table presents spanning tests results of regressing returns to the EIR trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{EIR} = \alpha + \sum_{k=1}^{6} \beta_{X_k} r_t^{X_k} + \sum_{j=1}^{6} \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor. The six most closely related anomalies, X, are Net Payout Yield, Growth in book equity, Share issuance (1 year), Change in equity to assets, Share issuance (5 year), Net equity financing. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.28	0.23	0.24	0.25	0.19	0.29	0.26
	[3.60]	[3.01]	[3.04]	[3.15]	[2.41]	[3.47]	[3.23]
Anomaly 1	20.23						17.33
	[6.65]						[4.24]
Anomaly 2		32.24					32.13
		[7.59]					[4.93]
Anomaly 3			17.48				-9.66
			[4.32]				[-1.81]
Anomaly 4				17.24			-6.74
				[4.32]			[-1.12]
Anomaly 5					23.63		17.10
					[5.57]		[3.23]
Anomaly 6						12.43	-6.78
						[3.24]	[-1.41]
mkt	-3.12	-5.19	-5.75	-6.55	-5.13	-4.79	-3.32
	[-1.63]	[-2.81]	[-3.03]	[-3.45]	[-2.72]	[-2.30]	[-1.63]
smb	-0.92	-5.26	-3.66	-4.40	-2.73	3.09	-1.79
	[-0.34]	[-1.97]	[-1.33]	[-1.61]	[-1.00]	[0.93]	[-0.55]
hml	4.55	8.64	12.26	10.31	14.54	11.48	5.38
	[1.19]	[2.39]	[3.34]	[2.78]	[3.99]	[3.02]	[1.39]
rmw	-16.52	-2.90	-14.08	-2.90	-13.13	-8.43	-7.88
	[-4.07]	[-0.80]	[-3.27]	[-0.78]	[-3.30]	[-1.87]	[-1.61]
cma	4.49	-10.20	10.38	3.82	9.87	7.41	-21.25
	[0.77]	[-1.52]	[1.75]	[0.57]	[1.73]	[1.20]	[-2.93]
umd	-0.36	-2.46	-2.70	-1.48	-3.45	-1.98	-1.92
	[-0.19]	[-1.34]	[-1.43]	[-0.79]	[-1.83]	[-1.01]	[-1.01]
# months	716	720	716	720	716	630	626
$\bar{R}^2$(%)	21	22	18	18	19	14	23

Panel A: Average rank

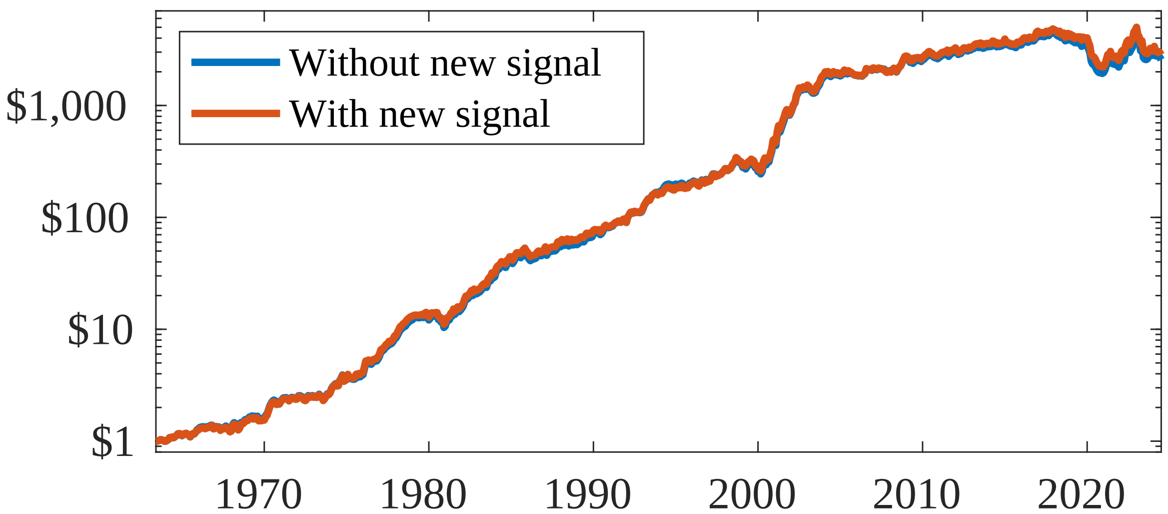


Figure 8: Combination strategy performance
This figure plots the growth of a $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as EIR. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

References

Balvers, R. J. and Zhang, D. (2017). Equity price booms and the real economy. *Journal of Financial Economics*, 124(3):486–503.

Carhart, M. M. (1997). On persistence in mutual fund performance. *Journal of Finance*, 52:57–82.

Chen, A. and Velikov, M. (2022). Zeroing in on the expected returns of anomalies. *Journal of Financial and Quantitative Analysis*, Forthcoming.

Chen, A. Y. and Zimmermann, T. (2022). Open source cross-sectional asset pricing. *Critical Finance Review*, 27(2):207–264.

Cochrane, J. H. and Piazzesi, M. (2005). Bond risk premia. *American Economic Review*, 95(1):138–160.

Daniel, K. and Titman, S. (2006). Market reactions to tangible and intangible information. *Journal of Finance*, 61(4):1605–1643.

Detzel, A., Novy-Marx, R., and Velikov, M. (2022). Model comparison with transaction costs. *Journal of Finance, Forthcoming*.

Fama, E. F. and French, K. R. (1993). Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, 33(1):3–56.

Fama, E. F. and French, K. R. (2015). A five-factor asset pricing model. *Journal of Financial Economics*, 116(1):1–22.

Fama, E. F. and French, K. R. (2018). Choosing factors. *Journal of Financial Economics*, 128(2):234–252.

Gomes, J. F. and Schmid, L. (2010). Levered returns. *Journal of Finance*, 65(2):467–494.

Kogan, L. and Papanikolaou, D. (2013). Firm characteristics and stock returns: The role of investment-specific shocks. *Review of Financial Studies*, 26(11):2718–2759.

Li, E. X., Livdan, D., and Zhang, L. (2009). Anomalies. *Review of Financial Studies*, 22(11):4301–4334.

Lyandres, E., Sun, L., and Zhang, L. (2008). The equity share in new issues and aggregate stock returns. *Journal of Finance*, 63(5):2279–2310.

McLean, R. D. and Pontiff, J. (2016). Does academic research destroy stock return predictability? *Journal of Finance*, 71(1):5–32.

Novy-Marx, R. and Velikov, M. (2016). A taxonomy of anomalies and their trading costs. *Review of Financial Studies*, 29(1):104–147.

Novy-Marx, R. and Velikov, M. (2023). Assaying anomalies. *Working paper*.

Pontiff, J. and Woodgate, A. (2008). Share issuance and cross-sectional returns. *Journal of Finance*, 63(2):921–945.

Zhang, L. (2005). The value premium. *Journal of Finance*, 60(1):67–103.